

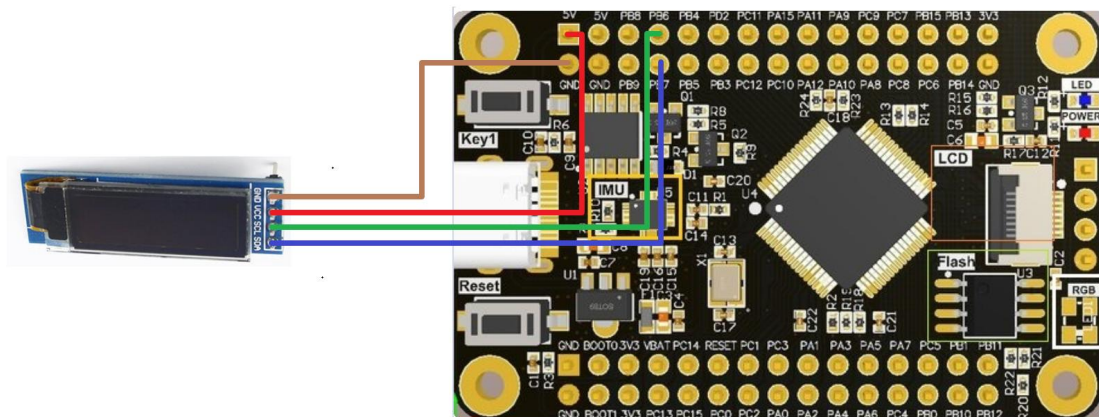
oled display string

一、 Learning objectives

In this course, we will learn how to use stm32f103c8t6 to control Oled display strings.

二、 Preparation before class

The hardware that needs to be used is several Dupont lines, stm32f103c8t6 development board, and 12832oled screen.



三、 Code Interpretation

1. Initialize the IIC pins PB6 and PB7 to be used

```

//初始化硬件IIC引脚 Initialize hardware IIC pins
void I2C_Configuration(void)
{
    GPIO_InitTypeDef GPIO_InitStructure;
    I2C_InitTypeDef I2C_InitStructure;

    RCC_APB1PeriphClockCmd(RCC_APB1Periph_I2C1, ENABLE);
    RCC_APB2PeriphClockCmd(RCC_APB2Periph_GPIOB, ENABLE);

    //PB6--SCL PB7--SDA
    GPIO_InitStructure.GPIO_Mode = GPIO_Mode_AF_OD;
    GPIO_InitStructure.GPIO_Pin = GPIO_Pin_6 | GPIO_Pin_7;
    GPIO_InitStructure.GPIO_Speed = GPIO_Speed_50MHz;

    GPIO_Init(GPIOB, &GPIO_InitStructure);

    I2C_DeInit(I2C1);
    I2C_InitStructure.I2C_Ack = I2C_Ack_Enable;
    I2C_InitStructure.I2C_AcknowledgedAddress = I2C_AcknowledgedAddress_7bit;
    I2C_InitStructure.I2C_ClockSpeed = 400000;
    I2C_InitStructure.I2C_DutyCycle = I2C_DutyCycle_2;
    I2C_InitStructure.I2C_Mode = I2C_Mode_I2C;
    I2C_InitStructure.I2C_OwnAddress1 = 0x30;

    I2C_Init(I2C1, &I2C_InitStructure);
    I2C_Cmd(I2C1, ENABLE);
}

```

2. String display function usage

```

//字符串显示函数 string display function
void OLED_ShowStr(unsigned char x,unsigned char y,unsigned char ch[],unsigned char TextSize)
{
    unsigned char c = 0,i = 0,j = 0;

    switch(TextSize)
    {
        case 1:

```

3. Main function

Initialize hardware gpio, delay, display string

```

int main(void)
{
    delay_init();//延迟函数初始化 delay function initialization
    OLED_Init();//oled初始化 oled initialization
    delay_ms(2000);
    OLED_Fill(0xFF);//填充白色 fill white
    delay_ms(2000);
    OLED_Fill(0x00);//黑屏 black screen
    delay_ms(2000);

    while(1)
    {
        OLED_ShowStr(0,0,"hello world!",2);//oled显示hello world! ! oled shows hello world! !
        OLED_ShowStr(0,3,"yahboom!",2);//oled显示yahboom ! oled shows yahboom!

    }
}

```

四、 Experimental phenomenon

After compiling the source file and uploading it to stm32, oled will display the string, as shown below

